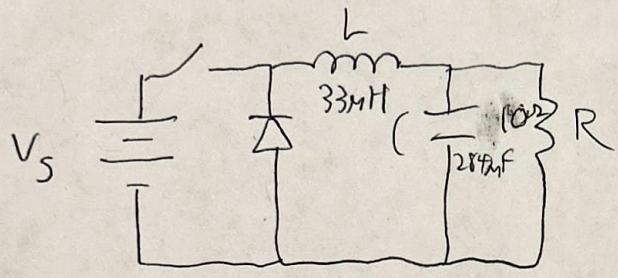
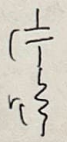
ECEN 164 Prelab 8

Buck converter:





← no r_C mentioned

transfer:

$$\frac{V_O}{D} = \frac{V_S}{LC}\,\frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$= \frac{V_S}{9.372\times10^{-9}s^2 + 3.3\times10^{-6}s + 1}$$

no zeros

poles: $\dfrac{-3.3\times10^{-6} \pm \sqrt{1.089\times10^{-11} - 3.7488\times10^{-8}}}{1.8744\times10^{-8}}$

$-176.056 + 10328.108j = 10329.608\angle 90.977$

and

$-176.056 - 10328.108j = 10329.608\angle -90.977$

two poles at same magnitude

frequencies

$\omega_0 = \frac{1}{\sqrt{LC}} = 10329.608$ ⟷ matches

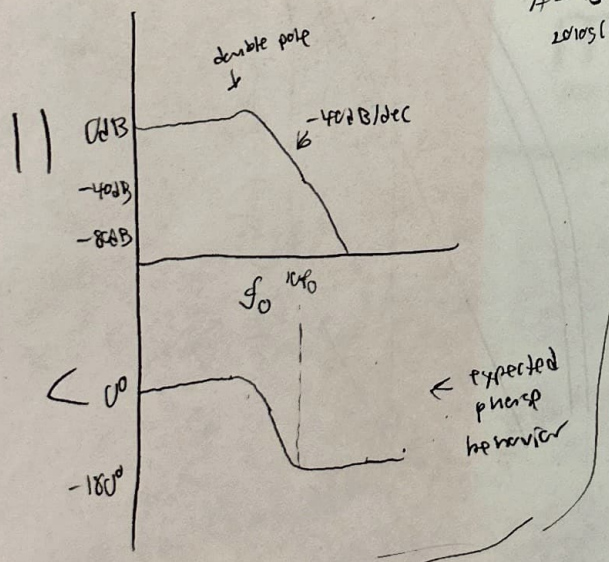
$f_0 = \frac{\omega_0}{2\pi} = 1644\,Hz$

no ESR

(say $V_S = 1$)

$A = V_S$

$20\log(1) = 0$



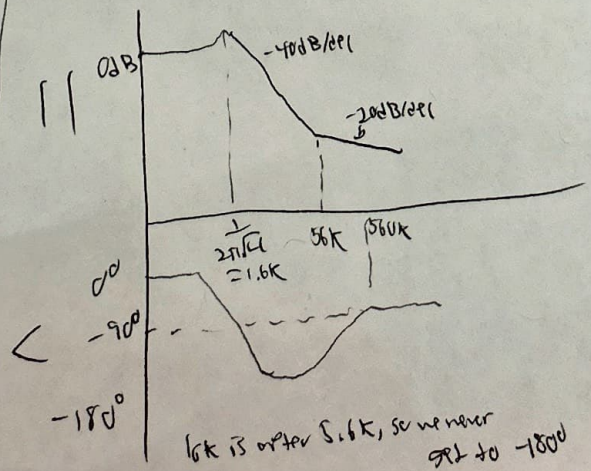
w/ ESR

$$\frac{V_O}{D} = \frac{V_S}{LC}\,\frac{1 + sr_C}{s^2 + s\left(\frac{1}{RC} + \frac{r_C}{L}\right) + \frac{1}{LC}}$$

$$= \frac{V_S}{9.372\times10^{-9}}\,\frac{1 + s2.84\times10^{-6}}{s^2 + s(655.143) + \frac{1}{9.372\times10^{-9}}}$$

$V_S = 1$ (again)

$f_{esr} = \frac{1}{2\pi r_C} = 56\,kHz$



16K is after 5.6K, so we never get to -180°

.16 ✓